



A systemic approach to explore the mechanisms of drug resistance and altered signaling cascades in extensively drug-resistant tuberculosis

S. Udhaya Kumar^a, Aisha Saleem^a, D. Thirumal Kumar^b,
V. Anu Preethi^c, Salma Younes^d, Hatem Zayed^e,
Iftikhar Aslam Tayubi^f, and C. George Priya Doss^{a,*}

^aSchool of Biosciences and Technology, Vellore Institute of Technology, Vellore, Tamil Nadu, India

^bDepartment of Bioinformatics, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, Tamil Nadu, India

^cSchool of Computer Science and Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu, India

^dTranslational Research Institute, Women's Wellness and Research Center, Hamad Medical Corporation, Doha, Qatar

^eDepartment of Biomedical Sciences, College of Health and Sciences, QU Health, Qatar University, Doha, Qatar

^fFaculty of Computing and Information Technology, King Abdul-Aziz University, Rabigh, Saudi Arabia

*Corresponding author: e-mail address: georgepriyadoss@vit.ac.in

Contents

1. Introduction	344
2. Materials and methods	346
2.1 Data procurement	346
2.2 Processing of data and screening for DEGs	347
2.3 Identification of PPI networks	347
2.4 Functional annotation and enrichment analysis	348
3. Results	348
3.1 Data procurement and screening for DEGs	348
3.2 Identification of PPI networks	351
3.3 Functional annotation and enrichment analysis	352
4. Discussion	357
5. Conclusion	360
Acknowledgments	361
Conflict of interest	361
Author contributions	361
Role of funding source	361
References	361

Abstract

Background and aim: The persistence of extensively drug-resistant (XDR) strains of *Mycobacterium tuberculosis* (MTB) continue to pose a significant challenge to the treatment and control of tuberculosis infections worldwide. XDR-MTB strains exhibit resistance against first-line anti-TB drugs, fluoroquinolones, and second-line injectable drugs. The mechanisms of drug resistance of MTB remains poorly understood. Our study aims at identifying the differentially expressed genes (DEGs), associated gene networks, and signaling cascades involved in rendering this pathogen resistant to multiple drugs, namely, isoniazid, rifampicin, and capreomycin. *Methods:* We used the microarray dataset GSE53843. The GEO2R tool was used to prioritize the most significant DEGs (top 250) of each drug exposure sample between XDR strains and non-resistant strains. The validation of the 250 DEGs was performed using volcano plots. Protein-protein interaction networks of the DEGs were created using STRING and Cytoscape tools, which helped decipher the relationship between these genes. The significant DEGs were functionally annotated using DAVID and ClueGO.

The concomitant biological processes (BP) and molecular functions (MF) were represented as dot plots. *Results and conclusion:* We identified relevant molecular pathways and biological processes, such as cell wall biogenesis, lipid metabolic process, ion transport, phosphopantetheine binding, and triglyceride lipase activity. These processes indicated the involvement of multiple interconnected mechanisms in drug resistance. Our study highlighted the impact of cell wall permeability, with the dysregulation of the mur family of proteins, as essential factors in the inference of resistance. Additionally, upregulation of genes responsible for ion transport such as *ctpF*, *arsC*, and *nark3*, emphasizes the importance of transport channels and efflux pumps in potentially driving out stress-inducing compounds. This study investigated the upregulation of the Lip family of proteins, which play a crucial role in triglyceride lipase activity. Thereby illuminating the potential role of drug-induced dormancy and subsequent resistance in the mycobacterial strains. Multiple mechanisms such as carboxylic acid metabolic process, NAD biosynthetic process, triglyceride lipase activity, phosphopantetheine binding, organic acid biosynthetic process, and growth of symbiont in host cell were observed to partake in resistance of XDR-MTB. This study ultimately provides a platform for important mapping targets for potential therapeutics against XDR-MTB.



1. Introduction

Mycobacterium tuberculosis (MTB) is the etiological agent of Tuberculosis (TB) (Arnold, 2007; Lillebaek et al., 2016). Around one-third of the global population has been infected by TB, causing over 60 million deaths since 2000, rendering it a cause of global concern (WHO, 2020). TB can exist in the lungs in the form of Pulmonary TB or extrapulmonary TB, which infect the lymph nodes, bones, abdomen, and meninges (Lee, 2015; Singh et al., 2020). Immunocompromised individuals are often more susceptible

to the disease (WHO, 2020). The prevalence of this disease has increased tremendously with the emergence of antibiotic resistance within the bacterium strains. Together with the alarming rate of occurrence of multi-drug resistance, the extended persistence of MTB infection enables the implementation of novel and successful anti-TB therapeutics, an international health concern (Karlas et al., 2010; Kumar et al., 2010). Individuals infected with non-resistant strains of the bacterium can be treated with standard first-line drugs such as isoniazid, rifampicin, ethambutol, and pyrazinamide (Luo et al., 2020; Singh et al., 2020). However, strains of TB have gained resistance to most of these drugs. Monoresistance is often exhibited against isoniazid and is the most common form of drug-resistant TB in the world (Chien et al., 2015; Furin, Cox, & Pai, 2019). Multidrug-resistant strains of TB (MDR-TB) harbor resistance against rifampicin and isoniazid. The existence of additional fluoroquinolone resistance and at least one injectable second-line drug (capreomycin, kanamycin, and amikacin) renders these MDR-TB strains drug-resistant (XDR-MTB) (Coll et al., 2018; Seung, Keshavjee, & Rich, 2015) extensively.

Various mechanisms have been attributed to the development of resistance. Growing evidence reveals that the sequential inclusion of genetic mutations in the genes that code for drug processing enzymes or drug targets are the main contributors to drug resistance acquisition (Coll et al., 2018). In resistant clinical TB isolates, a large number of mutations are being reported, most of which were located in the exon region, which implies that structural changes to the target protein of drugs play a crucial role in the resistance development (Sandgren et al., 2009; Zhang et al., 2013). Nevertheless, recognizing some resistance-related intergenic mutations also indicates a correlation between resistance and the regulation of gene expression (Zhang et al., 2013). It has been previously reported that resistance against isoniazid occurs due to mutations in the *katG* gene, responsible for converting the prodrug to active (INH). Pro-drug INH is activated upon forming an adduct with NAD. This adduct target the *inhA* protein, which plays an active role in cell wall synthesis. Mutations in *inhA* protein impart resistance to the bacterium as well (Hassan et al., 2020). Other prime contributors to drug resistance are cell envelope impermeability, cell wall lipid changes, which infer Rifampicin resistance, and overexpression of efflux pumps (Luo et al., 2020; Singh et al., 2020).

Mycobacterial virulence has been attributed to a wide range of virulence determinants and host responses at different pathogenesis stages (Madacki, Mas Fiol, & Brosch, 2019). To date, over 400 genes have been identified

as being essential for in vivo growth and survival of the bacterium (Zhang et al., 2013). Genes determined as potential virulent factors show immense diversity in their roles; for example, genes involved in primary cell metabolism, complex lipids of the cell envelope and components of ESX secretion systems are known to play equally significant roles in pathogenesis (Madacki et al., 2019). The convoluted pathways involved in the pathogenesis of MTB and mechanisms which infer drug resistance can be mapped via the identification of genetic determinants of the bacterium (Coll et al., 2018). It is the need of the hour to decipher the conundrum of drug resistance, which has rendered MTB a global threat. High throughput screening technologies such as microarray analysis can be employed to study diseases with complex pathogenesis (Alam et al., 2019; Cui & He, 2014; Kumar, Kumar, Siva, Doss, & Zayed, 2019; Mishra et al., 2020; Penn et al., 2018; Udhaya Kumar, Rajan, et al., 2020; Udhaya Kumar, Thirumal Kumar, Bithia et al., 2020; Udhaya Kumar, Thirumal Kumar, Siva, et al., 2020). In an attempt to ascertain the genetic determinants inferring drug resistance, we investigated the differentially expressed genes (DEGs) between non-resistant and extremely drug-resistant strains of MTB when treated with isoniazid, rifampicin, and capreomycin (GSE53843) (Guohua & Yu, 2014). This bioinformatics approach entailed retrieving DEGs data from the GEO database and its analysis using the web-tool GEO2R and visualization through the creation of volcano plots. Gene interaction networks were visualized using STRING and Cytoscape tools. Furthermore, protein interactions and pathways that are unique to drug resistance were explored to gain a better understanding of the complex mechanisms involved in extreme drug resistance.



2. Materials and methods

2.1 Data procurement

For this study, the dataset GSE53843 was acquired from NCBI's GEO database. Gene expression profiles of diverse experiments were stored in the GEO database and accessed by users for further analysis (Barrett et al., 2013). The keywords "*Mycobacterium tuberculosis*," "Drug-resistant," and "microarray" were utilized to extract related gene expression profile array datasets from the database. Samples of GSE53843 expression profiles comprise a reference strain H37Rv and two extensively drug-resistant (XDR) strains of MTB (Guohua & Yu, 2014). The three strains were treated with three anti-TB

drugs and water as control, resulting in a sample size of twelve. The platform GPL18119 Agilent-027618 *Mycobacterium tuberculosis* H37Rv 8X15K micro-array was employed to obtain the above sample set.

2.2 Processing of data and screening for DEGs

The statistical tool GEO2R was used to explore the DEGs upon different drug exposures between the XDR strains and the non-resistant reference strains. The GEO2R tool is incorporated with R/Bioconductor and Limma package (Ritchie et al., 2015). The *t*-test and Benjamini and Hochberg (False discovery rate), which are inbuilt statistical tools of GEO2R, aid in calculating *P*-value and FDR and play an essential role in determining the DEGs (Aubert, Bar-Hen, Daudin, & Robin, 2004). A series of four tests were carried out using GEO2R to extract DEGs on treatment with water (control), capreomycin, isoniazid, and rifampicin, resulting in four distinct datasets. We considered principal standard as $P \leq 0.05$ and $|\log(\text{fold change})| > 2$ to extract significant DEGs from the dataset. Upregulated genes were demarcated by a log fold change value of ≥ 2 and downregulated genes by $\log \text{FC} \leq -2$. The raw data and the top 250 DEGs were obtained from the dataset and processed via the removal of unidentified gene entries and hypothetical proteins. The selected DEGs were subjected to further investigation by constructing a volcano plot using RStudio and library Calibrate package, followed by identifying PPI networks.

2.3 Identification of PPI networks

Protein-protein interactions networks (PPIs) allow for the elucidation of relationships between proteins, which may be correlated through specific functional interactions (Childs & Larremore, 2021). In this study, we used STRING v.11 to identify the networks that existed among the DEGs. STRING is a web-based tool, which integrates information from multiple protein-protein association databases and renders computational predictions of these interactions in the form of networks (Szklarczyk et al., 2019). We eliminated interactions, which were inconsistent with the dataset by setting the cut off to an interaction score of >0.4 . We further visualized the networks obtained from STRING on Cytoscape software. The plug-in GeneMANIA was employed to obtain intersected clusters from the PPI network for the identified DEGs based on integrated scored values (Franz et al., 2018).

2.4 Functional annotation and enrichment analysis

To evaluate the functional determinants of the gene sets that have been significantly upregulated or downregulated during drug exposure, we performed an enrichment analysis using the DAVID v6.8. The Database for Annotation, Visualization, and Integrated Discovery (DAVID), incorporates information from a multitude of public bioinformatics databases and provides biological meaning to lists of genes (Huang et al., 2007). Using this tool, we determined the functional classification of GO terms and performed an enrichment analysis.

For comprehensive visualization and functional annotation of the Genes of GO terms and KEGG pathways, we utilized ClueGo software along with DAVID. ClueGo is a Cytoscape Plugin, which incorporates KEGG or BioCarta pathways and Gene Ontology (GO) terms to create a network of functionally grouped genes or pathways, representing the analysis results (Bindea et al., 2009). ClueGo v2.5.5 was used on the top 250 DEGs to obtain the GO terms and pathways involved in drug resistance against each of the three drugs.

To further elucidate the role of DEGs in the enriched biological pathways, dotplots were generated for each drug exposure dataset using GoPlot and ggplot2 R packages (Walter, Sánchez-Cabo, & Ricote, 2015; Wickham, 2016). DAVID database was utilized to functionally annotate the DEGs, whose regulatory levels in concomitant enriched pathways were visualized as dotplots.



3. Results

3.1 Data procurement and screening for DEGs

The dataset GSE53843 comprises gene expression profiles of two XDR strains of MTB and a non-resistant MTB strain as reference. Twelve samples were created due to the treatment of three strains with three anti-TB drugs and water as a control (Table 1). The GPL18119 platform (Agilent-027618 *Mycobacterium tuberculosis* H37Rv 8X15K microarray) was used in this study. The XDR strains treated with a specific anti-TB drug were compared against the reference strain treated with the same drug-using the NCBI-inbuilt GEO2R tool. The cut-off values of $|\log_{2}FC| \geq 2.0$ and P -values ≤ 0.05 for the DEGs were computed using the integrated limma package v3.26.8 and R/Bioconductor. The significant 250 DEGs between the reference strain and XDR strains were subsequently identified. The identified

DEGs from the XDR-MTB strains treated with water, capreomycin, isoniazid, rifampicin is provided in Supplementary material in the online version at <https://doi.org/10.1016/bs.apcsb.2021.02.002> in Supplementary Tables 1–4, respectively. Volcano plots created through R studio and calibrate (library) package was utilized to visualize these DEGs. Fig. 1 comprises the volcano plots of each of the three Anti TB drug-treated sets and water treated set as control. The significant genes satisfying all the statistical values ($P < 0.05$, $\logFC \geq 2$, and ≤ -2) are demarcated as green dots.

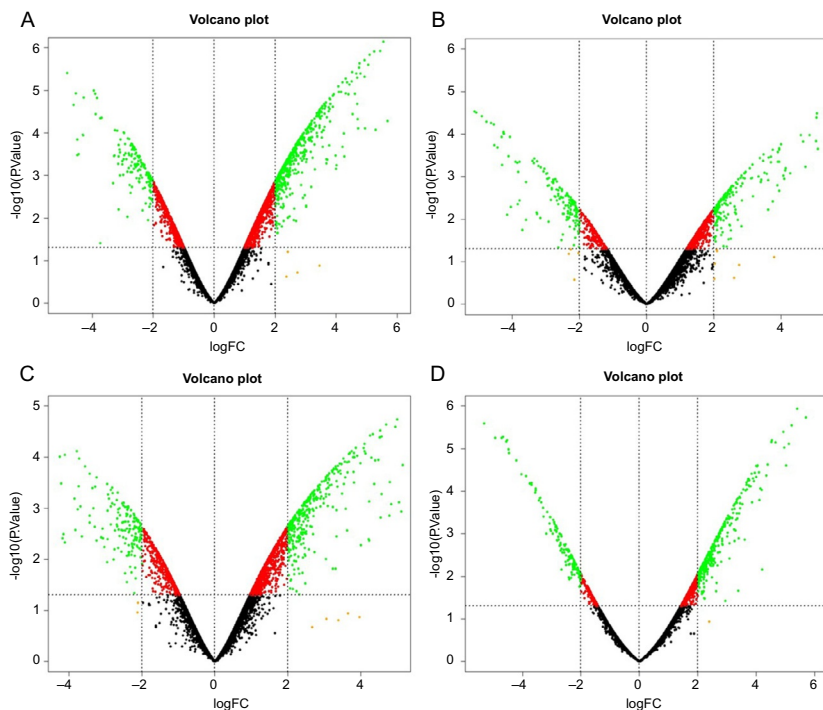


Fig. 1 Volcano plots comprising of DEGs visualized using R studio. Four plots were created, comparing the DEGs between control H37Rv strain and XDR strains treated with Water, capreomycin, isoniazid, and rifampicin. The legend is as follows: x-axis LogFC; y-axis $-\log_{10}$ of P -values. Upregulated genes are present in the right panel of the plot, and downregulated genes on the left. The red dots indicate P -values ≤ 0.05 and yellow dots indicate $\logFC \geq 2$ and $\logFC \leq -2$. The green dots signify DEGs that satisfy both $\logFC$ and P -value. The genes in the array that did not show any notable changes in expression are displayed as black dots.

Table 1 The MTB strains information from GSE53843 are provided.

Group	Accession	Title	Source name	Strain	Drug
1.	GSM1302015	H37Rv_CAP	1 × MIC CAP, 6 h	H37Rv	Capreomycin (CAP)
2.	GSM1302016	H37Rv_INH	10 × MIC INH, 6 h	H37Rv	Isoniazid (INH)
3.	GSM1302017	H37Rv_Ctrl	Water, 6 h	H37Rv	Water (control)
4.	GSM1302018	H37Rv_RIF	1 × MIC RIF, 6 h	H37Rv	Rifampicin (RIF)
5.	GSM1302019	XDR1219_CAP	1 × MIC CAP, 6 h	Extensively drug-resistant strain XDR1219	Capreomycin (CAP)
6.	GSM1302020	XDR1219_INH	10 × MIC INH, 6 h	Extensively drug-resistant strain XDR1219	Isoniazid (INH)
7.	GSM1302021	XDR1219_Ctrl	Water, 6 h	Extensively drug-resistant strain XDR1219	Water (control)
8.	GSM1302022	XDR1219_RIF	1 × MIC RIF, 6 h	Extensively drug-resistant strain XDR1219	Rifampicin (RIF)
9.	GSM1302023	XDR1221_CAP	1 × MIC CAP, 6 h	Extensively drug-resistant strain XDR1221	Capreomycin (CAP)
10.	GSM1302024	XDR1221_INH	10 × MIC INH, 6 h	Extensively drug-resistant strain XDR1221	Isoniazid (INH)
11.	GSM1302025	XDR1221_Ctrl	Water, 6 h	Extensively drug-resistant strain XDR1221	Water (control)
12.	GSM1302026	XDR1221_RIF	1 × MIC RIF, 6 h	Extensively drug-resistant strain XDR1221	Rifampicin (RIF)

3.2 Identification of PPI networks

A map depicting the functional and physical linkages among the DEGs’ proteins was created using STRING and Cytoscape tools. Initial data for PPIs was procured in the tabular text through STRING and then further visualized using Cytoscape v3.7.1. Fig. 2 comprises the PPI networks of DEGs obtained through water-treated strains and capreomycin indicated by blue and pink nodes, respectively. As depicted in the figure, we found 49 nodes and 63 edges for water-treated strains, whereas, for capreomycin-treated strains, we found 34 nodes and 49 edges. Fig. 3 depicts the PPI network of isoniazid and rifampicin exposures in green and purple nodes, respectively. As a result, we found 33 nodes and 30 edges for Isoniazid-treated

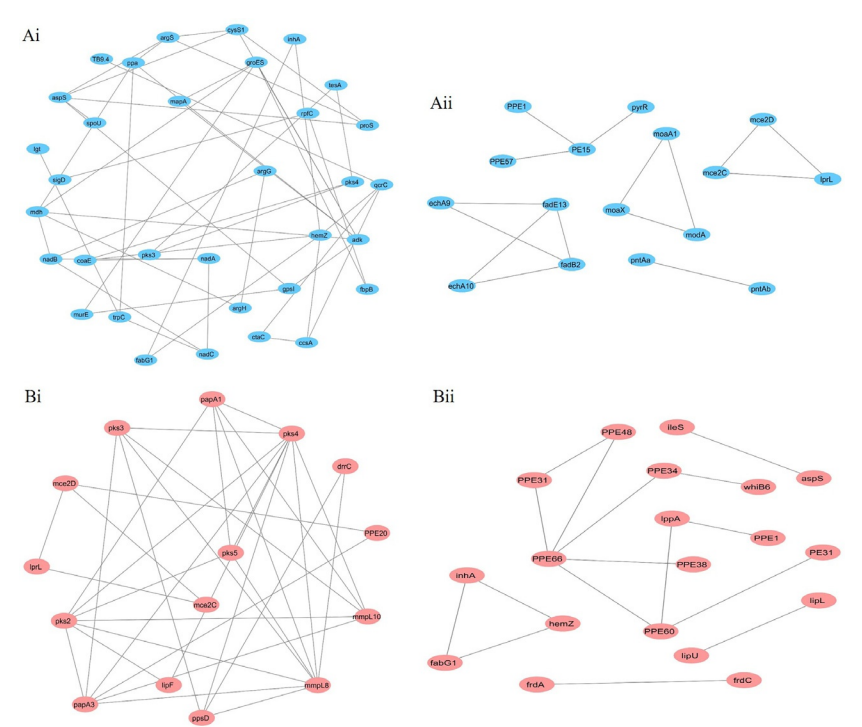


Fig. 2 Interactions between DEGs of GSE53843 dataset portrayed through PPI networks. STRING tool was used to retrieve the nodes and edges, which were then, plotted using Cytoscape software. Blue nodes (Ai and Aii) represent the PPI networks of DEGs of water-treated strains, pink nodes represent DEGs of strains treated with capreomycin (Bi and Bii). Edges are displayed in gray.

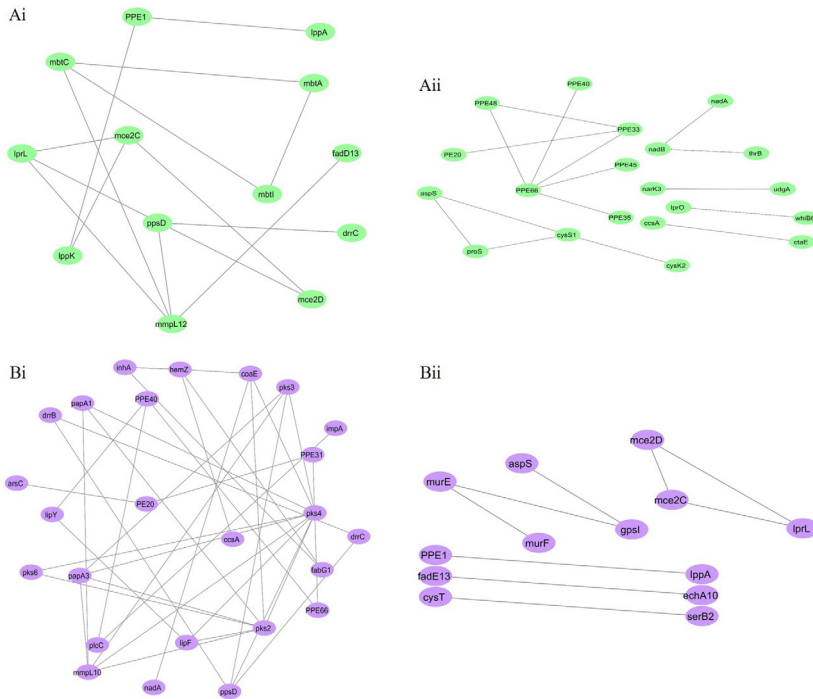


Fig. 3 PPI networks mapping the interactions between DEGs of the GSE53843 dataset. Cytoscape software was used to plot the nodes and edges retrieved from the STRING tool. Green nodes (Ai and Aii) represent the PPI networks of DEGs of isoniazid-treated strains. Purple nodes represent the DEGs of strains treated with rifampicin (Bi and Bii). Edges are displayed in gray.

strains, whereas, for rifampicin-treated strains, we found 38 nodes and 46 edges. The number of proteins are denoted as nodes, and edges represent their interactions.

3.3 Functional annotation and enrichment analysis

The functional enrichment analysis for the DEGs from the dataset was carried out using the Cytoscape plug-in ClueGO. The MF and BP terms of the GO annotation of each drug treatment was set, i.e., control, capreomycin, isoniazid, and rifampicin, which are depicted in Figs. 4–7, respectively. A hypergeometric test, which is two-sided with Benjamini-Hochberg correction, kappa score > 0.4, and $P < 0.05$ as a significant condition, determined the statistical options for the ClueGO enrichment annotation. The DEG's PPI network in the absence of drug exposure (Fig. 4) was enriched

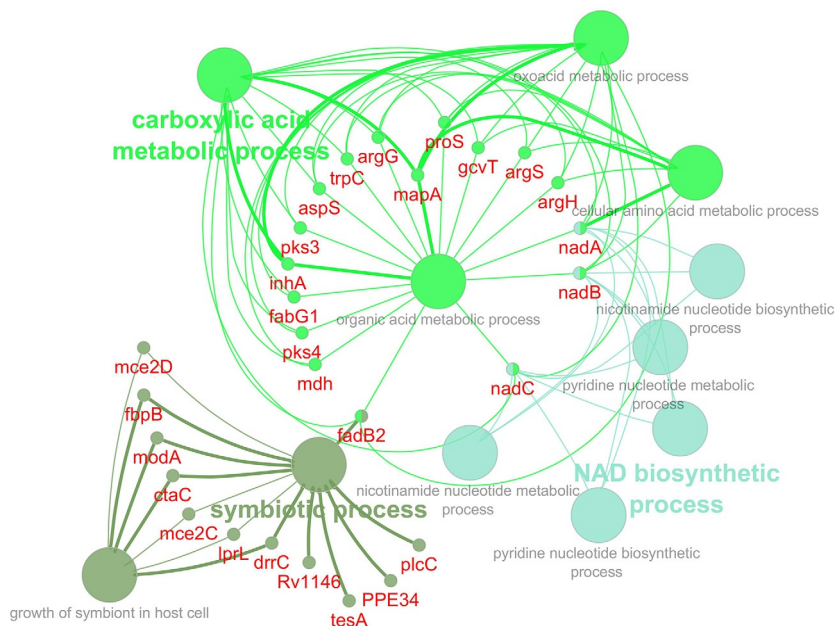


Fig. 4 The ClueGO Cytoscape plugin was used to visualize the enrichment analysis of strains treated with water by Gene Ontology (GO) terms. The visualization tool incorporates KEGG and Reactome pathways, which allow it to display the significant Molecular Functions (MF) and Biological Processes (BP) exhibited by the top 250 DEGs. The node size is dictated by P -value ≤ 0.05 and node colors display the distinct yet interconnected MF and BP groups enriched upon treatment. The most important functional GO terms which denominate the MF and BP groups are indicated in bold font. Red font is used to display the names of the DEGs in each group.

in BP such as symbiotic processes (GO:0044403), NAD biosynthetic process (GO:0009435), and carboxylic acid metabolic process (GO:0019752). MF and BP of strains exposed to capreomycin (Fig. 5) were significantly enriched in phosphopantetheine binding (GO:0031177), sulfur compound biosynthetic process (GO:0044272), and triglyceride lipase activity (GO:0004806). For the enrichment analysis of isoniazid exposure (Fig. 6), MF and BP were predominantly enriched in Iron-sulfur cluster binding (GO:0051536), ion transport (GO:0006811), and organic acid biosynthetic process (GO:0016053). Rifampicin treatment (Fig. 7) exhibited enrichment of DEGs involved in transport (GO:0006810), growth of symbiont in a host cell (GO:0044119), and phosphopantetheine binding (GO:0031177). Additionally, GO terms enriched during INH treatment illustrated as dotplots (Fig. 8A) comprised of oxoacid metabolic process (GO:0043436), organonitrogen compound

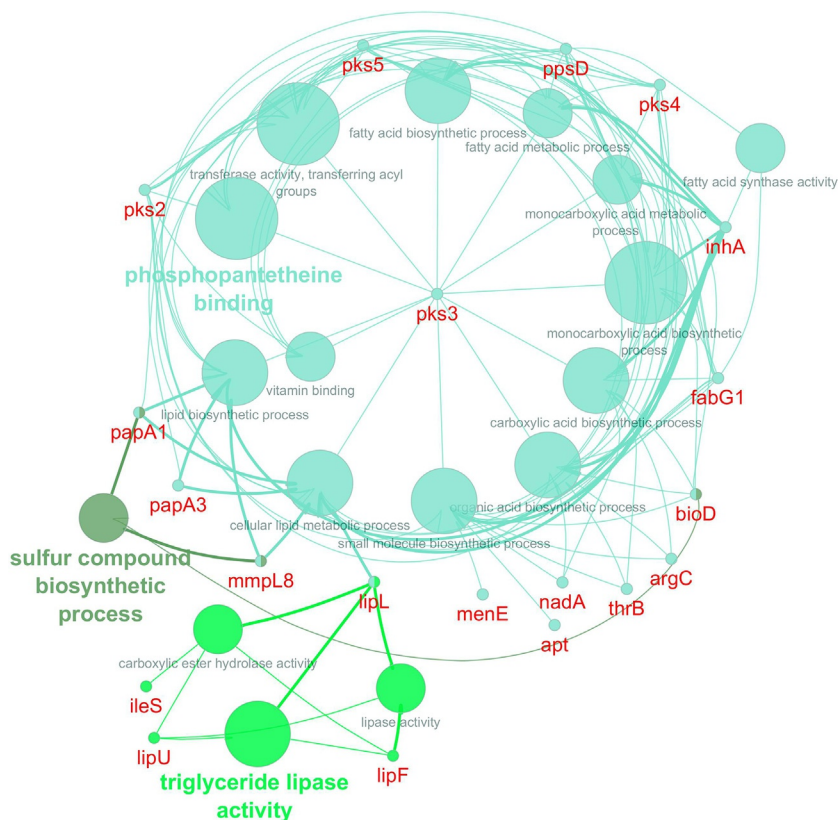


Fig. 5 Enrichment analysis of capreomycin-treated strains using ClueGO plugin of Cytoscape software. A kappa score of 0.4 determines the connectivity of the network between distinct pathways. A P -value $\leq$ of 0.5 is required for the pathways to be deemed significant in the enrichment analysis. Node size depends upon the P -value and node color specifies the functional class they belong to. The salient GO/pathway terms defining the MF and BP groups are displayed in bold font. Red font signifies the DEGs involved in each group. Triglyceride lipase activity, phosphopantetheine, and sulfur compound biosynthetic processes are significantly enriched among strains treated with capreomycin.

biosynthetic process (GO:1901566), organic acid metabolic process (GO:0006082), organic acid biosynthetic process (GO:0016053), monocarboxylic acid biosynthetic process (GO:0072330), carboxylic acid biosynthetic process (GO:0046394), and carboxylic acid metabolic process (GO:0019752). For RIF treatment, the enriched terms were involved in lipid metabolic process (GO:0006629), lipid biosynthetic process (GO:0008610), peptidoglycan-based cell wall biogenesis (GO:0009273), cell wall biogenesis (GO:0042546),

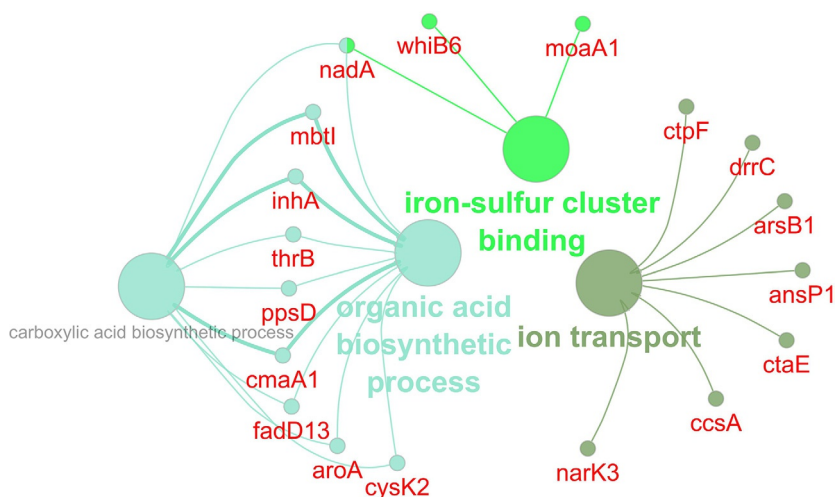


Fig. 6 Enrichment analysis of isoniazid-treated strains using ClueGO plugin of Cytoscape software. The plugin acquires data from KEGG and REACTOME pathways as well and therefore provides a detailed enrichment analysis of the DEGs. Iron-sulfur cluster binding, ion transport and organic acid biosynthetic processes are indicated to play a critical role in the inference of isoniazid resistance in XDR-MTB strains. Red font indicates DEGs involved in the metabolic processes. Significant physiological processes are indicated in bold font. Node size and node color differ with *P*-value and associated functional class, respectively.

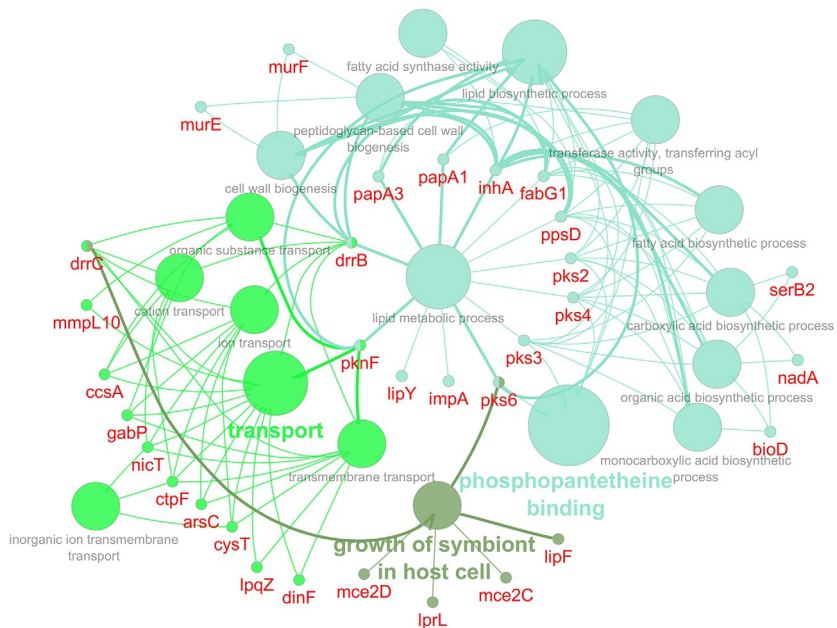


Fig. 7 Enrichment map comprising Go terms and pathways involved in rifampicin resistance visualized using ClueGO plugin of Cytoscape software. Transport mechanisms, growth of symbiont in host cells and phosphopantetheine binding mechanisms are the significantly enriched physiological processes among strains treated with rifampicin. Node size is determined by a *P*-value $\leq$ of 0.05 various MF, and BP are attributed to node colors. The names of the significant GO terms, which define the enriched MF and BP groups, are displayed in bold font. DEGs comprising these groups are denominated in red font.

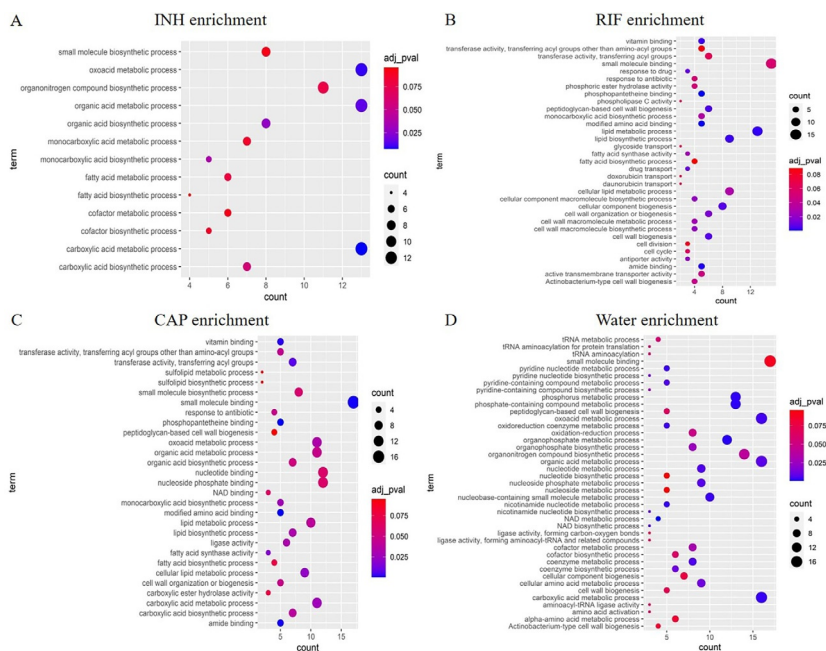


Fig. 8 Graphical representation of enriched GO terms in the form of a DotPlot. The GO terms of BP and MF are plotted against the count of genes. The dots size represents the number of genes among the significant DEGs associated with the GO term. The color of the dots represent their significance in accordance with the adjusted *P*-value. (A) enriched GO terms on exposure to isoniazid, (B) enriched Go terms on exposure to rifampicin, (C) enriched Go terms on exposure to capreomycin, (D) enriched Go terms in control-water.

phosphopantetheine binding (GO:0031177), fatty acid synthase activity (GO:0004312) and small-molecule binding (GO:0036094) (Fig. 8B). In CAP treatment, some of the standard enriched terms were carboxylic acid biosynthetic process (GO:0046394) and organic acid biosynthetic process (GO:0016053). Cellular lipid metabolic process (GO:0044255), carboxylic acid metabolic process (GO:0019752), lipid biosynthetic process (GO:0008610), oxoacid metabolic process (GO:0043436), lipid metabolic process (GO:0006629), and organic acid metabolic process (GO:0006082) were the additional enriched BP terms and phosphopantetheine binding (GO:0031177), modified amino acid binding (GO:0072341), amide binding (GO:0033218), and small molecule binding (GO:0036094) were the enriched molecular functions (Fig. 8C). The standard terms enriched in XDR-MTB during treatment with water were carboxylic acid metabolic process

(GO:0019752), pyridine nucleotide metabolic process (GO:0019362), NAD biosynthetic process (GO:0009435), nicotinamide nucleotide biosynthetic process (GO:0019359), and cellular amino acid metabolic process (GO:0006520). Organophosphate metabolic process (GO:0090407), phosphorus metabolic process (GO:0006793), oxoacid metabolic process (GO:0043436), organic acid metabolic process (GO:0006082), and organonitrogen compound biosynthetic process (GO:1901566) were identified as significantly enriched GO terms as well (Fig. 8D). Through the above results, it can be proclaimed that the XDR strains have adapted diverse mechanisms to overcome lethal dosages of anti-MTB drugs, and thereby, have gained resistance against them.



4. Discussion

Drug resistance in TB is often attributed to genetic mutations, which affect either the bacterial enzymes that activate prodrugs or the drug target itself. However, there exist multiple alternate mechanisms, which facilitate drug resistance in this pathogenic bacterium (Peñuelas-Urquides, Castorena-Torres, SilvaRamírez, & de León, 2017). Transcriptome analysis elucidated through comparison of DEGs, evaluation of pathways associated with these DEGs, and interactions between these processes may give us an insight into the overall mechanism of drug resistance in extremely drug-resistant strains of MTB (Li et al., 2020; Peñuelas-Urquides et al., 2017). In the present study, we investigated gene expression profiles of dataset GSE53843 to identify the interconnected processes and pathways involved in drug resistance. The dataset comprised of microarray results obtained through the exposure of XDR strains of MTB and MTB reference strain to water and inhibitory concentrations of the anti-TB drugs; isoniazid, rifampicin, and capreomycin, which were then scrutinized through the identification of DEGs, evaluation of protein interaction networks, functional annotation, and enrichment of MF, BP and pathways involved.

Rifampicin's known mechanism of bacterial growth inhibition is through the suppression of RNA synthesis by binding of the drug to β sub-unit of RNA polymerase, RpoB. Resistance to a lethal concentration of rifampicin has been attributed to mutations in the *rpoB* gene (Peñuelas-Urquides, et al., 2017). However, in a recent study carried out by Zhu et al. (2018), bacterial populations exhibiting resistance upon exposure to an inhibitory concentration of rifampicin were upregulating the expression

of gene *rpoB* (Zhu et al., 2018). This conforms with the results of our study as well, in which *rpoB* is upregulated in XDR-MTB strains that are exposed to rifampicin. Another factor known to contribute towards rifampicin resistance is the cell wall, a crucial drug target (Luo et al., 2020). ClueGo enrichment analysis identified GO:0009273 peptidoglycan-based cell wall biogenesis, GO:0042546 cell wall biogenesis, and cell wall-related pathways such as GO:0006629. Lipid metabolic process GO:0008610 and lipid biosynthetic process as significantly enriched processes among the strains exposed to rifampicin. Genes *murF* and *murE*, which are considered essential drug targets due to their catalytic activity in peptidoglycan synthesis, were significantly down-regulated upon rifampicin exposure; however, upregulation of *murA* gene, which plays a role in cell wall synthesis (Luo et al., 2020), was observed (Fig. 7). It is important to note that the development of low cell permeability is an intrinsic mechanism of drug resistance (Louw et al., 2009); the dys-regulation of cell wall-related genes observed in our study may corroborate this mechanistic property. The bactericidal effect of rifampicin has recently been associated with sequential hydroxyl radical formation (Piccaro, Pietraforte, Giannoni, Mustazzolu, & Fattorini, 2014), resulting in the bacterium's lyses. Upregulation of universal stress protein (*uspB*) was observed in this study with INH treated MTB (with logFC value 3) (see Table 3 in Supplementary material in the online version at <https://doi.org/10.1016/bs.apcsb.2021.02.002>), which complements the view that bactericidal drugs increase oxidative stress in MTB present within macrophages, and therefore may further validate bactericidal effects of the drug (Piccaro et al., 2014). Additionally, UspB is an integral part of the ATP binding ABC transporter protein, which is essential for the growth of the bacterium in vitro (Fullam, Prokes, Fütterer, & Besra, 2016).

The activation of pro-drug isoniazid occurs through the catalase-peroxidase activity of an enzyme encoded by *katG*, which target genes involved in mycolic acid synthesis pathway, *inhA* (enoyl ACP reductase) and *kasA* (beta-ketoacyl ACP synthase) (Vilchèze & Jacobs, 2014). A total of 5053 genes were differentially expressed in XDR-MTB strains exposed to a lethal concentration of INH. The key enriched terms were ion transport GO:0006811, iron-sulfur cluster binding GO:0051536, organic acid biosynthetic process GO:0016053 and carboxylic acid biosynthetic process GO:0046394 (Fig. 6). Significant upregulation of gene *inhA* was observed in CAP, INH, and RIF treated MTB (with logFC value >3); however, there was no notable change in the expression of *katG*. A study carried out by Ando et al. (2011) suggested that the downregulation of *katG* is a contributing factor

for isoniazid resistance (Ando et al., 2011); similarly, *inhA* upregulation has also been attributed toward increased isoniazid resistance (Vilchèze & Jacobs, 2014). Isoniazid is a hydrazide derivative of a carboxylic acid, and *Mycobacterium tuberculosis* is known to be highly sensitive to weak acids (Maresca, Vullo, Scozzafava, Manole, & Supuran, 2013). Some of the enriched terms in isoniazid treatment were carboxylic acid metabolic processes such as GO:0019752, oxoacid metabolic process GO:0043436, organic acid metabolic process GO:0006082, organic acid biosynthetic process GO:0016053, monocarboxylic acid biosynthetic process GO:0072330, and carboxylic acid biosynthetic process GO:0046394, which are indicative of the mechanisms involved in overriding this selective pressure.

Transporters and Efflux pumps play a vital role in developing multidrug resistance via partaking in the extrusion of specific compounds and even seemingly unrelated compounds (Louw et al., 2009). DEGs involved in ion transport GO:0006811 were observed in both rifampicin exposure as well as isoniazid exposure. Genes responsible for calcium pumping (*ctpF*) during oxidative stress conditions (Maya-Hoyos, Rosales, Novoa-Aponte, Castillo, & Soto, 2019), nitrite extrusion (*nark3*), and arsenic transport (*arsC*) were observed to be significantly upregulated in response to isoniazid and rifampicin exposures, throwing light upon the involvement of drug metabolism and transport in the development of multidrug resistance.

Upon treatment with capreomycin, a total of 5053 genes were differentially expressed, which were primarily involved in phosphopantetheine binding GO:0031177, sulfur compound biosynthetic process GO:0044272 and triglyceride lipase activity GO:0004806 (Fig. 5). Although previous studies have identified ribosomes to be the target of CAP, the proteins that interact with this drug are not yet known (Lin et al., 2014). However, studies have investigated the processes of lipase activity and fatty acid synthesis in contributing to virulence factors and determinants of resistance in MTB. Stress factors induce the production of lipid bodies in the organism, known to bring about a state of dormancy in-vivo and consequently confer resistance (Deb et al., 2009). To enter this state of dormancy, MTB stores energy in the form of triacylglycerols (Deb et al., 2009). Significant involvement of gene *lipU* was observed in our study (Fig. 5), which is known to hydrolyze short-chain carbon substrates (Li et al., 2017). Similarly, *lipY* carries out hydrolysis of long-chain stored triglycerides, leading to stored energy utilization (Deb et al., 2009). In our study, the genes *lipF*, *lipL*, and *lipU* were observed to play essential roles in triglyceride lipase activity, indicating the possibility of acquiring dormancy upon capreomycin exposure (Fig. 5).

The enrichment of Phosphopantetheine GO:0031177 binding was observed in both rifampicin exposure and capreomycin treatment (Figs. 5 and 7). Phosphopantetheine adenylyltransferase (PPAT) plays a crucial role in the CoA biosynthesis pathway (Wubben & Mesecar, 2011). The DEGs involved in the enriched Phosphopantetheine binding process belonged to the polyketide synthase encoding genes (*pks*). *Pks* genes decorate the cell envelope with unique lipid polyketides, which are critical in virulence and add to the cell envelope's complexity (Parvez et al., 2018). We observed a significant downregulation of these genes upon rifampicin and capreomycin exposure, which indicates a crucial role of this process and associated cell envelope proteins in developing resistance. MTB is known to possess multiple virulence factors and, consequently, multiple mechanisms to evade anti-TB drugs (Madacki et al., 2019). Our study highlighted some of the mechanisms that played vital roles in the XDR strain's survival upon drug exposure. The key factors that were identified as contributors to drug resistance were modifications in the cell wall and its related components, upregulation of universal stress proteins, upregulation of transport channels, metabolism of stress-inducing components such as carboxylic acid derivatives, and transition to the stage of dormancy via the production of lipid storage molecules and subsequent energy utilization through hydrolysis of these molecules.



5. Conclusion

Our study investigated drug resistance mechanisms in XDR-MTB strains, specifically against Capreomycin, isoniazid, and rifampicin, using the microarray dataset GSE53843. Upon analyses of the DEGs present between the drug-exposed control strains, we identified multiple factors as contributors to drug resistance in XDR-MTB strains, some of which were in accordance with previously identified determinants involvement of cell wall-associated proteins, transport channels, and dysregulation of known drug targets. The peculiar findings of this study, which require further speculation, are processes involved in the stress-induced response, such as in the presence of carboxylic acids, the importance of phosphopantetheine binding in mechanistic drug resistance, and the acquirement of dormancy upon capreomycin exposure. These observations although peculiar and exciting, require functional studies to identify their appropriate role in the drug-induced response and possible role in the acquirement of resistance. Ultimately, our findings provide a platform for mapping the numerous

interconnected mechanisms involved in drug resistance and distinguishing novel therapeutic targets for the treatment of XDR-MTB.

Acknowledgments

Mr. Udhaya Kumar. S, one of the authors, gratefully acknowledges the Indian Council of Medical Research (ICMR), India, for providing him a Senior Research Fellowship [ISRM/11(93)/2019]. The authors would like to take this opportunity to thank the management of VIT for providing the necessary facilities and encouragement to carry out this work.

Conflict of interest

The authors have declared that no conflicts of interest exist.

Author contributions

SUK and CGPD were involved in the design of the study. SUK was involved in the data collection and experiment. SUK, AS, DTK, VAP, SY, IAT, and HZ involved in the acquisition, analysis, and interpreting of the results. SUK and AS drafted the manuscript. CGPD supervised the entire study and was involved in study design, the acquisition, analysis, understanding of the data, and critically reviewed the manuscript. All authors edited and approved the submitted version of the article.

Role of funding source

No funding agency was involved in the present study.

References

- Alam, A., Imam, N., Ahmed, M. M., Tazyeen, S., Tamkeen, N., Farooqui, A., et al. (2019). Identification and classification of differentially expressed genes and network meta-analysis reveals potential molecular signatures associated with tuberculosis. *Frontiers in Genetics*, 10, 932. <https://doi.org/10.3389/fgene.2019.00932>.
- Ando, H., Kitao, T., Miyoshi-Akiyama, T., Kato, S., Mori, T., & Kirikae, T. (2011). Downregulation of katG expression is associated with isoniazid resistance in *Mycobacterium tuberculosis*. *Molecular Microbiology*, 79(6), 1615–1628. <https://doi.org/10.1111/j.1365-2958.2011.07547.x>.
- Arnold, C. (2007). Molecular evolution of *Mycobacterium tuberculosis*. *Clinical Microbiology and Infection*, 13(2), 120–128. <https://doi.org/10.1111/j.1469-0691.2006.01637.x>.
- Aubert, J., Bar-Hen, A., Daudin, J. J., & Robin, S. (2004). Determination of the differentially expressed genes in microarray experiments using local FDR. *BMC Bioinformatics*, 5, 125. <https://doi.org/10.1186/1471-2105-5-125>.
- Barrett, T., Wilhite, S. E., Ledoux, P., Evangelista, C., Kim, I. F., Tomashevsky, M., et al. (2013). NCBI GEO: Archive for functional genomics data sets—Update. *Nucleic Acids Research*, 41(Database issue), D991–D995. <https://doi.org/10.1093/nar/gks1193>.
- Bindea, G., Mlecnik, B., Hackl, H., Charoentong, P., Tosolini, M., Kirilovsky, A., et al. (2009). ClueGO: A cytoscape plug-in to decipher functionally grouped gene ontology and pathway annotation networks. *Bioinformatics*, 25(8), 1091–1093. <https://doi.org/10.1093/bioinformatics/btp101>.

- Chien, J.-Y., Chen, Y.-T., Wu, S.-G., Lee, J.-J., Wang, J.-Y., & Yu, C.-J. (2015). Treatment outcome of patients with isoniazid mono-resistant tuberculosis. *Clinical Microbiology and Infection*, 21(1), 59–68. <https://doi.org/10.1016/j.cmi.2014.08.008>.
- Childs, L. M., & Larremore, D. B. (2021). Network models for malaria: Antigens, dynamics, and evolution over space and time. In O. Wolkenhauer (Ed.), *Systems medicine* (pp. 277–294). Academic Press. <https://doi.org/10.1016/B978-0-12-801238-3.11512-0>.
- Coll, F., Phelan, J., Hill-Cawthorne, G. A., Nair, M. B., Mallard, K., Ali, S., et al. (2018). Genome-wide analysis of multi- and extensively drug-resistant *Mycobacterium tuberculosis*. *Nature Genetics*, 50(2), 307–316. <https://doi.org/10.1038/s41588-017-0029-0>.
- Cui, T., & He, Z.-G. (2014). Improved understanding of pathogenesis from protein interactions in *Mycobacterium tuberculosis*. *Expert Review of Proteomics*, 11(6), 745–755. <https://doi.org/10.1586/14789450.2014.971762>.
- Deb, C., Lee, C.-M., Dubey, V. S., Daniel, J., Abomoelak, B., Sirakova, T. D., et al. (2009). A novel in vitro multiple-stress dormancy model for *Mycobacterium tuberculosis* generates a lipid-loaded, drug-tolerant, dormant pathogen. *PLoS One*, 4(6), e6077. <https://doi.org/10.1371/journal.pone.0006077>.
- Franz, M., Rodriguez, H., Lopes, C., Zuberi, K., Montojo, J., Bader, G. D., et al. (2018). GeneMANIA update 2018. *Nucleic Acids Research*, 46(W1), W60–W64. <https://doi.org/10.1093/nar/gky311>.
- Fullam, E., Prokes, I., Fütterer, K., & Besra, G. S. (2016). Structural and functional analysis of the solute-binding protein UspC from *Mycobacterium tuberculosis* that is specific for amino sugars. *Open Biology*, 6(6), 160105. <https://doi.org/10.1098/rsob.160105>.
- Furin, J., Cox, H., & Pai, M. (2019). Tuberculosis. *The Lancet*, 393(10181), 1642–1656. [https://doi.org/10.1016/S0140-6736\(19\)30308-3](https://doi.org/10.1016/S0140-6736(19)30308-3).
- Guohua, Y., & Yu, L. (2014). *Expression profiling of extensively drug-resistant tuberculosis isolates and H37Rv exposed to anti-TB compounds*. GEO Accession Viewer. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE53843>.
- Hassan, S., Sudhakar, V., Mary, M. B. N., Babu, R., Doble, M., Dadar, M., et al. (2020). Computational approach identifies protein off-targets for isoniazid-NAD adduct: Hypothesizing a possible drug resistance mechanism in *Mycobacterium tuberculosis*. *Journal of Biomolecular Structure and Dynamics*, 38(6), 1697–1710. <https://doi.org/10.1080/07391102.2019.1615987>.
- Huang, D. W., Sherman, B. T., Tan, Q., Kir, J., Liu, D., Bryant, D., et al. (2007). DAVID bioinformatics resources: Expanded annotation database and novel algorithms to better extract biology from large gene lists. *Nucleic Acids Research*, 35(Web Server issue), W169–W175. <https://doi.org/10.1093/nar/gkm415>.
- Karlas, A., Machuy, N., Shin, Y., Pleissner, K.-P., Artarini, A., Heuer, D., et al. (2010). Genome-wide RNAi screen identifies human host factors crucial for influenza virus replication. *Nature*, 463(7282), 818–822. <https://doi.org/10.1038/nature08760>.
- Kumar, S. U., Kumar, D. T., Siva, R., Doss, C. G. P., & Zayed, H. (2019). Integrative bioinformatics approaches to map potential novel genes and pathways involved in ovarian cancer. *Frontiers in Bioengineering and Biotechnology*, 7, 391. <https://doi.org/10.3389/fbioe.2019.00391>.
- Kumar, D., Nath, L., Kamal, M. A., Varshney, A., Jain, A., Singh, S., et al. (2010). Genome-wide analysis of the host intracellular network that regulates survival of *Mycobacterium tuberculosis*. *Cell*, 140(5), 731–743. <https://doi.org/10.1016/j.cell.2010.02.012>.
- Lee, J. Y. (2015). Diagnosis and treatment of extrapulmonary tuberculosis. *Tuberculosis Respiratory Disease*, 78(2), 47–55. <https://doi.org/10.4046/trd.2015.78.2.47>.
- Li, H., Chen, T., Yu, L., Guo, H., Chen, L., Chen, Y., et al. (2020). Genome-wide DNA methylation and transcriptome and proteome changes in *Mycobacterium tuberculosis* with Para-aminosalicylic acid resistance. *Chemical Biology & Drug Design*, 95(1), 104–112. <https://doi.org/10.1111/cbdd.13625>.

- Li, C., Li, Q., Zhang, Y., Gong, Z., Ren, S., Li, P., et al. (2017). Characterization and function of *Mycobacterium tuberculosis* H37Rv lipase Rv1076 (LipU). *Microbiological Research*, 196, 7–16. <https://doi.org/10.1016/j.micres.2016.12.005>.
- Lillebaek, T., Norman, A., Rasmussen, E. M., Marvig, R. L., Folkvardsen, D. B., Andersen, Å. B., et al. (2016). Substantial molecular evolution and mutation rates in prolonged latent *Mycobacterium tuberculosis* infection in humans. *International Journal of Medical Microbiology*, 306(7), 580–585. <https://doi.org/10.1016/j.ijmm.2016.05.017>.
- Lin, Y., Li, Y., Zhu, N., Han, Y., Jiang, W., Wang, Y., et al. (2014). The antituberculosis antibiotic capreomycin inhibits protein synthesis by disrupting interaction between ribosomal proteins L12 and L10. *Antimicrobial Agents and Chemotherapy*, 58(4), 2038–2044. <https://doi.org/10.1128/AAC.02394-13>.
- Louw, G. E., Warren, R. M., van Pittius, N. C. G., McEvoy, C. R. E., Helden, P. D. V., & Victor, T. C. (2009). A balancing act: efflux/influx in mycobacterial drug resistance. *Antimicrobial Agents and Chemotherapy*, 53(8), 3181–3189. <https://doi.org/10.1128/AAC.01577-08>.
- Luo, X., Pan, J., Meng, Q., Huang, J., Wang, W., Zhang, N., et al. (2020). High-throughput screen for cell wall synthesis network module in *Mycobacterium tuberculosis* based on integrated bioinformatics strategy. *Frontiers in Bioengineering and Biotechnology*, 8, 607. <https://doi.org/10.3389/fbioe.2020.00607>.
- Madacki, J., Mas Fiol, G., & Brosch, R. (2019). Update on the virulence factors of the obligate pathogen *Mycobacterium tuberculosis* and related tuberculosis-causing mycobacteria. *Infection, Genetics and Evolution*, 72, 67–77. <https://doi.org/10.1016/j.meegid.2018.12.013>.
- Maresca, A., Vullo, D., Scozzafava, A., Manole, G., & Supuran, C. T. (2013). Inhibition of the β -class carbonic anhydrases from *Mycobacterium tuberculosis* with carboxylic acids. *Journal of Enzyme Inhibition and Medicinal Chemistry*, 28(2), 392–396. <https://doi.org/10.3109/14756366.2011.650168>.
- Maya-Hoyos, M., Rosales, C., Novoa-Aponte, L., Castillo, E., & Soto, C. Y. (2019). The P-type ATPase CtpF is a plasma membrane transporter mediating calcium efflux in *Mycobacterium tuberculosis* cells. *Heliyon*, 5(11), e02852. <https://doi.org/10.1016/j.heliyon.2019.e02852>.
- Mishra, S., Shah, M. I., Udhaya Kumar, S., Thirumal Kumar, D., Gopalakrishnan, C., Al-Subaie, A. M., et al. (2020). Network analysis of transcriptomics data for the prediction and prioritization of membrane-associated biomarkers for idiopathic pulmonary fibrosis (IPF) by bioinformatics approach. *Advances in Protein Chemistry and Structural Biology*, 120, 349–377. Academic Press <https://doi.org/10.1016/bs.apcsb.2020.10.003>.
- Parvez, A., Giri, S., Giri, G. R., Kumari, M., Bisht, R., & Saxena, P. (2018). Novel type III polyketide synthases biosynthesize methylated polyketides in *Mycobacterium marinum*. *Scientific Reports*, 8(1), 6529. <https://doi.org/10.1038/s41598-018-24980-1>.
- Penn, B. H., Netter, Z., Johnson, J. R., Von Dollen, J., Jang, G. M., Johnson, T., et al. (2018). An Mtb-human protein-protein interaction map identifies a switch between host antiviral and antibacterial responses. *Molecular Cell*, 71(4), 637–648.e5. <https://doi.org/10.1016/j.molcel.2018.07.010>.
- Peñuelas-Urquides, K., Castorena-Torres, F., SilvaRamírez, B., & de León, M. B. (2017). Drug resistance in *Mycobacterium tuberculosis*. *Mycobacterium—Research and Development*, 117–129. <https://doi.org/10.5772/intechopen.69656>.
- Piccaro, G., Pietraforte, D., Giannoni, F., Mustazzolu, A., & Fattorini, L. (2014). Rifampin induces hydroxyl radical formation in *Mycobacterium tuberculosis*. *Antimicrobial Agents and Chemotherapy*, 58(12), 7527–7533. <https://doi.org/10.1128/AAC.03169-14>.
- Ritchie, M. E., Phipson, B., Wu, D., Hu, Y., Law, C. W., Shi, W., et al. (2015). Limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research*, 43(7), e47. <https://doi.org/10.1093/nar/gkv007>.

- Sandgren, A., Strong, M., Muthukrishnan, P., Weiner, B. K., Church, G. M., & Murray, M. B. (2009). Tuberculosis drug resistance mutation database. *PLoS Medicine*, 6(2), e1000002. <https://doi.org/10.1371/journal.pmed.1000002>.
- Seung, K. J., Keshavjee, S., & Rich, M. L. (2015). Multidrug-resistant tuberculosis and extensively drug-resistant tuberculosis. *Cold Spring Harbor Perspectives in Medicine*, 5(9), a017863. <https://doi.org/10.1101/cshperspect.a017863>.
- Singh, R., Dwivedi, S. P., Gaharwar, U. S., Meena, R., Rajamani, P., & Prasad, T. (2020). Recent updates on drug resistance in Mycobacterium tuberculosis. *Journal of Applied Microbiology*, 128(6), 1547–1567. <https://doi.org/10.1111/jam.14478>.
- Szklarczyk, D., Gable, A. L., Lyon, D., Junge, A., Wyder, S., Huerta-Cepas, J., et al. (2019). STRING v11: Protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Research*, 47(D1), D607–D613. <https://doi.org/10.1093/nar/gky1131>.
- Udhaya Kumar, S., Rajan, B., Thirumal Kumar, D., Preethi, V. A., Abunada, T., Younes, S., et al. (2020). Involvement of essential signaling cascades and analysis of gene networks in diabetes. *Genes*, 11(11), 1256. <https://doi.org/10.3390/genes11111256>.
- Udhaya Kumar, S., Thirumal Kumar, D., Bithia, R., Sankar, S., Magesh, R., Sidenna, M., et al. (2020). Analysis of differentially expressed genes and molecular pathways in familial hypercholesterolemia involved in atherosclerosis: A systematic and bioinformatics approach. *Frontiers in Genetics*, 11, 734. <https://doi.org/10.3389/fgene.2020.00734>.
- Udhaya Kumar, S., Thirumal Kumar, D., Siva, R., George Priya Doss, C., Younes, S., Younes, N., et al. (2020). Dysregulation of signaling pathways due to differentially expressed genes from the B-cell transcriptomes of systemic lupus erythematosus patients—A bioinformatics approach. *Frontiers in Bioengineering and Biotechnology*, 8, 276. <https://doi.org/10.3389/fbioe.2020.00276>.
- Vilchèze, C., & Jacobs, W. R., Jr. (2014). Resistance to isoniazid and ethionamide in Mycobacterium tuberculosis: Genes, mutations, and causalities. *Microbiology Spectrum*, 2(4), 1–21. <https://doi.org/10.1128/microbiolspec.MGM2-0014-2013>.
- Walter, W., Sánchez-Cabo, F., & Ricote, M. (2015). GOpot: An R package for visually combining expression data with functional analysis. *Bioinformatics*, 31(17), 2912–2914. <https://doi.org/10.1093/bioinformatics/btv300>.
- WHO. (2020). *Global tuberculosis report*. World Health Organization. <https://www.who.int/teams/global-tuberculosis-programme/data>.
- Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis* (2nd ed.). Springer International Publishing. <https://doi.org/10.1007/978-3-319-24277-4>.
- Wubben, T., & Mesecar, A. D. (2011). Structure of Mycobacterium tuberculosis phosphopantetheine adenylyltransferase in complex with the feedback inhibitor CoA reveals only one active-site conformation. *Acta Crystallographica Section F: Structural Biology and Crystallization Communications*, 67(Pt 5), 541–545. <https://doi.org/10.1107/S1744309111010761>.
- Zhang, Y. J., Reddy, M. C., Ioerger, T. R., Rothchild, A. C., Dartois, V., Schuster, B. M., et al. (2013). Tryptophan biosynthesis protects mycobacteria from CD4 T-cell-mediated killing. *Cell*, 155(6), 1296–1308. <https://doi.org/10.1016/j.cell.2013.10.045>.
- Zhu, J.-H., Wang, B.-W., Pan, M., Zeng, Y.-N., Rego, H., & Javid, B. (2018). Rifampicin can induce antibiotic tolerance in mycobacteria via paradoxical changes in rpoB transcription. *Nature Communications*, 9(1), 4218. <https://doi.org/10.1038/s41467-018-06667-3>.